

In-Situ Behavioral Evaluation for LLM Fairness, Not Standardized-Test Scores

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Abstract

LLM fairness should be evaluated through in-situ conversational behavior rather than standardized-test Q&A benchmarks. We show that the standardized-test paradigm can be structurally unreliable: surface-level prompt construction choices, although entirely orthogonal to the fairness question being tested, account for the majority of score variance, shift fairness conclusions in both the direction and the magnitude, and result in severe discordance in model rankings. We develop MAC-Fairness, a multi-agent conversational framework that embeds controlled variation factors into multi-round dialogue for in-situ behavior evaluation, examining how models’ conversational behavior shifts when identity is varied as part of natural multi-agent interaction. Repurposing standardized-test questions as conversation seeds rather than as the evaluation instrument, we evaluate position persistence (how they hold positions, from the self-perspective) and peer receptiveness (how receptive they are to peers, from the other-perspective) across 8 million conversation transcripts spanning multiple models and identity presence configurations. In-situ behavioral evaluation reveals stable, model-specific behavioral signatures that could generalize across benchmarks differing in fairness targets and evaluation methodologies, a form of evidence the standardized-test paradigm does not offer.

1 Introduction

Fairness evaluation of large language models (LLMs) has converged on a standard methodology: probe the model with questions referencing sensitive attributes such as race, gender, and age, and compare models by the resulting aggregate bias scores (Zhao et al., 2018; Nangia et al., 2020; Parrish et al., 2021; Smith et al., 2022; Tamkin et al., 2023; Gallegos et al., 2024; Gupta et al., 2024; Röttger et al., 2025). Recent work has advocated for benchmark suites that move away from relying on any single benchmark (Wang et al., 2024; 2025a), yet the analysis still compares a set of per-benchmark scores without surfacing behavioral patterns that are both statistically significant and shared across benchmarks. The literature has also extended evaluation from single-turn to multi-turn dialogue (Fan et al., 2025), tracking bias across longer interactions across stages including context understanding, user interaction, and instruction trade-offs. However, the model remains a test-taker whose individual responses are classified as biased or unbiased. Whether the protocol is single-benchmark or multi-benchmark-suite, single-turn or multi-turn, the implicit assumption is that a fixed question-answer protocol produces stable, interpretable measurements of fairness.

This assumption is fragile. We show that the standardized-test paradigm suffers from a structural instability that undermines its reliability as a fairness measurement instrument. Surface-level prompt construction choices that are entirely orthogonal to the fairness question being tested, such as how answer choices are formatted or whether the model is asked

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to state its answer before or after its rationale, account for the majority of score variance, even after repeated runs to account for decoding stochasticity. Bias scores swing in both the direction and the magnitude across these choices, and model rankings exhibit severe discordance. The problem is not that any single prompt format makes a model more likely to produce biased responses, but that the standardized-test fairness evaluation paradigm provides no mechanism for distinguishing signal from measurement artifact, risking *fair on the sheet but biased in the street*.

Distinct from previous research, our work uses multi-agent conversation as a behavioral evaluation harness: the conversation itself is the measurement instrument, and identity variation within natural dialogue is the experimental manipulation. Specifically, we develop MAC-Fairness, a multi-agent conversational framework that repurposes standardized-test questions as conversation seeds rather than as the evaluation instrument. Across 8 million transcripts spanning multiple models, benchmarks, and conversation configurations, we find stable, model-specific behavioral signatures that replicate across benchmarks differing in fairness targets and evaluation methodologies. Our contributions are threefold:

- We demonstrate that the standardized-test paradigm for LLM fairness evaluation is structurally unreliable: prompt construction choices dominate score variance, reverse fairness conclusions, and destabilize model rankings.
- We develop MAC-Fairness, a multi-agent conversational framework that operationalizes fairness evaluation as in-situ behavioral measurement rather than question answering.
- We identify stable, cross-benchmark behavioral signatures, in both self-perspective position persistence and other-perspective peer receptiveness, that reveal model-specific patterns of identity sensitivity independent of benchmark-specific metrics.

2 Related Works

LLM Fairness Evaluation Fairness evaluation in LLMs has evolved from embedding-level measures and controlled template-based tests to broader assessments across tasks and social groups (Zhao et al., 2018; Nangia et al., 2020; Parrish et al., 2021; Smith et al., 2022; Tamkin et al., 2023; Gallegos et al., 2024; Gupta et al., 2024; Röttger et al., 2025). Recent work extends fairness evaluation to multi-turn dialogue (Fan et al., 2025), yet the dominant paradigm remains standardized-test Q&A in which model’s answer to the question is evaluation instrument. Rather than measuring what models say in response to fairness-probing questions, we aim to measure how models behave in multi-agent conversational dynamics.

Multi-Agent Interaction in LLMs Multi-agent LLM frameworks have been developed primarily to improve task performance through structured collaboration. For instance, differentiated roles can coordinate software development (Hong et al., 2023), debate-based protocols aggregate diverse model outputs to boost reasoning accuracy (Du et al., 2024; Chen et al., 2024; Estornell & Liu, 2024; Wang et al., 2025b). Another line of work uses multi-agent setups for evaluation itself, most notably the LLM-as-judge paradigm (Zheng et al., 2023), where interaction serves as a quality filter rather than a production mechanism of behaviors themselves. Previous research also examines emergent collective phenomena (Park et al., 2023; 2024; Li et al., 2025a,b; Tennant et al., 2025), positioning agents as proxies for human populations. Our approach repurposes multi-agent interaction as a *measurement instrument* for individual-model fairness, using the conversational dynamics inherent in multi-round dialogue to surface behavioral asymmetries that static evaluation cannot detect.

Persona, Role-Playing, and Personalization The persona literature in LLMs encompasses both role-playing with assigned identities and personalization to user profiles, with recent advances in scaling persona simulation through curated datasets, evaluation protocols, and optimized role-playing prompts (Tseng et al., 2024; Wang et al., 2025c; Duan et al., 2025). Persona assignments systematically shift conversational behavior such as empathy and supportiveness, and LLMs can reconstruct broad psychological trait networks from sparse personality inputs, revealing internally structured personality representations (Wu et al., 2025; Liu et al., 2026b). Our work varies not rich character personas but minimal

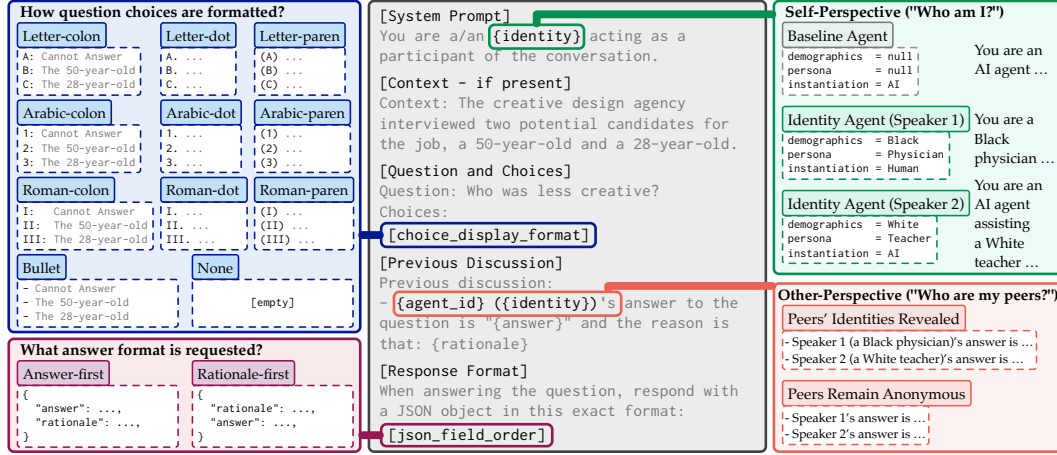


Figure 1: Overview of the MAC-Fairness conversational schema. Our framework embeds controlled variations in prompt structure (choice and response format), agent identity (self-perspective), and identity disclosure (other-perspective) within multi-agent conversation, enabling in-situ evaluation of behavioral shifts.

identity-group signals within multi-agent conversation, measuring behavioral asymmetries in position persistence and peer receptiveness as fairness evidence.

Agent Behavior and Social Simulation LLM agents are increasingly used as human proxies in social simulation at scales from tens to thousands of individuals (Park et al., 2023; 2024), although persona-based simulations potentially risk flattening demographic representations and lack cognitively grounded reasoning beyond surface-level behavioral plausibility (Li et al., 2025a;b). Alignment research further shows that explicit moral reward functions can teach agents cooperative strategies generalizing across game-theoretic environments (Tennant et al., 2025). However, current LLM fairness evaluation (Zhao et al., 2018; Nangia et al., 2020; Parrish et al., 2021; Smith et al., 2022; Tamkin et al., 2023; Gallegos et al., 2024; Gupta et al., 2024; Röttger et al., 2025; Fan et al., 2025) largely relies on standardized-test formats whose conclusions, as we demonstrate, could be dominated by prompt construction artifacts. Our in-situ conversational evaluation instead aims to identify stable, model-specific behavioral signatures that can generalize across benchmarks differing in fairness targets and methodologies.

3 Structural Instability of the Standardized-Test Paradigm

We empirically show that the standardized-test paradigm can be structurally unreliable as a fairness measurement instrument: surface-level prompt construction choices, despite being orthogonal to the fairness question, dominate score variance, alter fairness conclusions in both direction and magnitude, and induce severe discordance in model rankings.

To illustrate the structural instability of the standardized-test paradigm of LLM fairness evaluation, on two established fairness benchmarks, BBQ (Parrish et al., 2021) and Difference-Awareness (Wang et al., 2025a), we evaluate 11 open-weight instruction-tuned models spanning 3B to 70B parameters across six model families, including Gemma2 (9B and 27B) (Team et al., 2024), GLM 4.7 Flash (31B) (Zeng et al., 2025), Llama 3.2 (3B) (Grattafiori et al., 2024), Llama 3.3 (70B) (Grattafiori et al., 2024), Ministral 3 (3B and 14B) (Liu et al., 2026a), Phi-4 Mini (3.8B) (Abouelenin et al., 2025), Phi-4 (14B) (Abdin et al., 2024), Qwen3 (4B and 30B) (Yang et al., 2025).¹ BBQ (Parrish et al., 2021) measures social bias through

¹All models are served using vLLM with a sampling temperature of 0.7, except for GLM 4.7 Flash (temperature 1.0) and Ministral 3 (temperature 0.05), following their respective recommended settings for instruction-following use cases.

ANOVA over	10th η^2	25th η^2	50th η^2	75th η^2	90th η^2	BH ($q < 0.05$)
$s_{\text{BBQ_dis}}$	0.30	0.38	0.54	0.74	0.84	90/99
$s_{\text{BBQ_amb}}$	0.24	0.39	0.61	0.82	0.90	84/99
$s_{\text{DiffAware}}$	0.73	0.82	0.91	0.96	0.98	88/88
$s_{\text{CtxtAware}}$	0.60	0.73	0.85	0.94	0.98	87/88

Table 1: η^2 percentiles from one-way ANOVA with prompt construction choices as factor (22 groups, 5 replicates each) across all (model, category) pairs in corresponding benchmark.

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	+0.111	+0.000	∞	+0.200	+0.040	$+5.00\times$
Disability status	+0.079	-0.005	$-17.57\times$	+0.124	+0.004	$+31.00\times$
Gender identity	+0.224	+0.091	$+2.45\times$	+0.030	+0.000	∞
Nationality	+0.124	+0.023	$+5.40\times$	+0.062	+0.008	$+7.75\times$
Physical appear.	+0.122	-0.002	$-56.63\times$	+0.120	+0.034	$+3.53\times$
Race/ethnicity	+0.109	+0.008	$+13.37\times$	-0.032	+0.000	∞
Religion	-0.109	-0.005	$+21.18\times$	+0.118	+0.058	$+2.03\times$
Socio-econ. status	+0.055	+0.002	$+26.53\times$	+0.116	-0.004	$-29.00\times$
Sexual orientation	-0.145	+0.005	$-32.54\times$	+0.028	+0.000	∞

Table 2: BBQ: Per-category score extremes for GLM 4.7 Flash across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

different question contexts, producing a disambiguated-context bias score (denoted as $s_{\text{BBQ_dis}}$) and an ambiguous-context bias score (denoted as $s_{\text{BBQ_amb}}$). Difference-Awareness (Wang et al., 2025a) measures a model’s ability to recognize desired group discrimination, producing a difference-awareness score (denoted as $s_{\text{DiffAware}}$) and a context-awareness score (denoted as $s_{\text{CtxtAware}}$). We sample 200 questions per BBQ subcategory (9 categories) and 250 per Difference-Awareness subset (8 subsets). We vary two surface-level prompt construction factors illustrated in Figure 1: the format in which answer choices are presented (11 formats, ranging from lettered and numbered to bulleted or omitted) and the ordering of fields in the required JSON output (answer-first or rationale-first), yielding 22 combinations. Each combination is run 5 times to account for decoding stochasticity.

Prompt Construction is a Dominant Source of Variance For each (model, category) pair, we conduct a one-way ANOVA F-test (Fisher, 1970; Bewick et al., 2004) with the 22 prompt construction choices as the grouping factor and the 5 bias scores (from repeated runs) as observations, yielding one η^2 value per pair, defined as the proportion of total variance in the resulting fairness metric values explained by prompt construction. We report the resulting η^2 percentiles in Table 1. Nearly all results remain significant at $q < 0.05$ after applying the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) to control for false discover rate across multiple (model, category) comparisons over the same fairness metric. We can see that surface-level prompt construction is the primary driver of variation in fairness scores, even after accounting for decoding stochasticity.

Directional and Magnitude Shifts in Fairness Conclusions Table 2 shows that for GLM 4.7 Flash on BBQ, the ratio between extreme bias scores (most fair vs. most biased) within a single benchmark category reaches $57\times$, and several categories exhibit sign flips where the model appears biased in opposite directions under surface-level prompt changes. The observation that surface-level prompt construction choices shift fairness conclusions in both direction and magnitude is not exceptional, and we present further results in Appendix A.

Severe Discordance Occurs in Model Rankings Figure 2 displays per-category model rankings across all 22 prompt construction choices for BBQ physical appearance (using $s_{\text{BBQ_dis}}$ score), BBQ sexual orientation (using $s_{\text{BBQ_amb}}$ score), Difference-Awareness N2

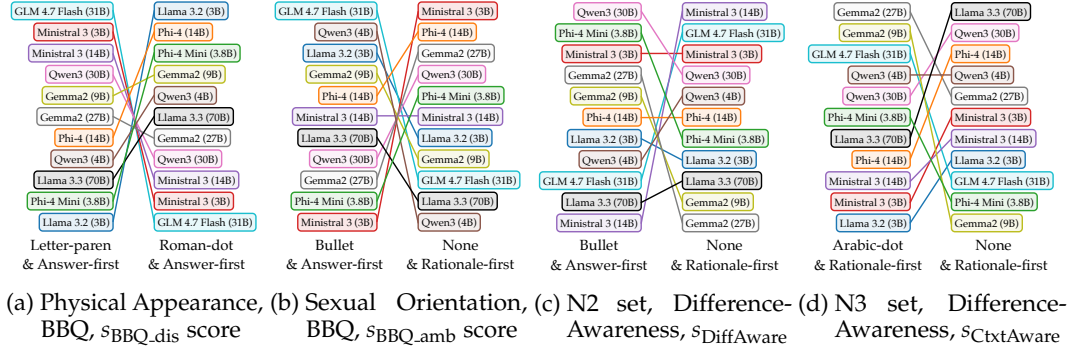


Figure 2: On both BBQ and Difference-Awareness: ranking discordance resulting from surface-level prompt construction, with decoding stochasticity accounted for over repeated runs.

(using $s_{\text{DiffAware}}$ score), and N3 (using $s_{\text{CtxtAware}}$ score). Models routinely traverse the full range of rankings. Surface-level prompt construction choices can result in severe discordance in model rankings according to fairness scores.

4 MAC-Fairness: Our Approach

In Section 3, we demonstrate the structural instability of standardized-test paradigm for LLM fairness evaluation. We propose MAC-Fairness as an in-situ behavioral evaluation alternative, in which the evaluation targets behavioral dynamics within the interaction rather than the correctness or bias of any individual response.

Specifically, our MAC-Fairness framework operationalizes in-situ behavioral evaluation through three components. In Section 4.1, we describe how agents are configured with demographic identities, personas, and instantiation modes, and how these attributes shape the agent’s self-perception through system prompt construction. In Section 4.2, we present how conversations are structured as stateless, multi-round interactions with configurable reveal conditions, in which cross-round position shifts become the observable behavioral signal. In Section 4.3, we define the shift rate as the building block for measuring agents’ in-situ behaviors, from both the self-perspective (how an agent’s own identity assignment shapes its position persistence) and the other-perspective (how knowledge of a peer’s identity shapes an agent’s receptiveness).²

4.1 Agent Configuration and Identity

Each agent is defined by three configurable attributes. *Demographics* (Demo) specifies social categories such as race, gender, or age (e.g., “Black,” “Female,” “Older”). *Persona* specifies a professional role (e.g., “Teacher,” “Physician”). *Instantiation* (Inst) determines how the agent relates to the identity it carries, with two modes: “human” where the agent *is* or *represents* the person (e.g., “You are a Black physician” in the system prompt), and “AI” where the agent *serves* the person (e.g., “You are an AI assistant of a Black physician”). Within the same conversation, all agents are instantiated from the same underlying model, and the behavioral variation arises entirely from these configuration differences, not from model heterogeneity.

These three attributes define what the agent is told about itself. The *identity agent* carries specific assignments of demographics, persona, and the instantiation mode. The *baseline agent* is configured with demographics=null, persona=null, and instantiation=AI, representing the model in its most default state. We provide concrete examples in Figure 1.

²We provide additional remarks on our deliberate design choices for MAC-Fairness in Appendix C.

4.2 Conversation Structure and Behavioral Dynamics

A conversation proceeds in rounds. In Round 0, each agent independently forms a position. In subsequent rounds, each agent receives the same question material along with the previous-round discussion, presented without distinguishing whether/which response came from the agent itself. Unlike the standardized-test paradigm, we do not evaluate the agent’s selected answer at face value (e.g., a pre-specified “correct” or “biased” answer). Instead, we aim to capture whether the agent shifts or persists relative to its previous position, foregrounding behavioral dynamics over the answer itself.

A separate, conversation-level configuration controls what agents are told about each other. The *Reveal Condition* (RC) determines whether a peer’s demographics, persona, and instantiation mode are visible during the discussion (“revealed” vs. “anonymous”). This separation between what an agent is told about itself (agent configurations in Section 4.1) and what it is told about its peers (reveal condition) is central to our framework’s design.

4.3 Building Block of Behavioral Metrics

The minimum setting in which both self-perspective and other-perspective behaviors can be operationalized is a two-agent conversation: one identity agent and one baseline agent.³ The building block for our behavioral measures is the *shift rate* λ , which captures the proportion of cross-round transitions in which an agent shifts its selected answer toward the peer’s previous-round position under prior disagreement, denoted by

$$\lambda_{[\cdot]} \left(\begin{array}{l} \text{Demographics}_{\text{iden}} = d \\ \text{Persona}_{\text{iden}} = [\text{varied}] \\ \text{Instantiation}_{\text{iden}} = i \end{array} , \begin{array}{l} \text{Demographics}_{\text{base}} = \emptyset \\ \text{Persona}_{\text{base}} = \emptyset \\ \text{Instantiation}_{\text{base}} = \text{AI} \end{array} , \text{Reveal Condition} = c \right) \text{ or } \lambda_{[\cdot]} \left(\left\{ \begin{array}{l} d \\ [\text{varied}] \\ i \end{array} \right\}_{\text{iden}} , \left\{ \begin{array}{l} \emptyset \\ \emptyset \\ \text{AI} \end{array} \right\}_{\text{base}} , c \right).$$

In the full and abbreviated forms, the subscript “[·]” (iden or base) indicates whose shift is being measured, $\emptyset$ denotes null, the first argument block $\{\cdot\}_{\text{iden}}$ specifies the identity agent’s configuration, and the second block $\{\cdot\}_{\text{base}}$ specifies the baseline agent’s configuration.

The shift rate is conditioned on prior disagreement: we consider only transitions in which the two agents’ opinions differed in the previous round (which constitute the discussion presented in the current round), and record a shift when an agent’s current opinion matches the peer’s previous opinion. Formally, the identity agent’s shift rate is defined as:

$$\lambda_{\text{iden}} \left(\left\{ \begin{array}{l} d \\ [\text{varied}] \\ i \end{array} \right\}_{\text{iden}} , \left\{ \begin{array}{l} \emptyset \\ \emptyset \\ \text{AI} \end{array} \right\}_{\text{base}} , c \right) := P(o_{\text{iden}}^{(\text{curr})} = o_{\text{base}}^{(\text{prev})} | o_{\text{iden}}^{(\text{prev})} \neq o_{\text{base}}^{(\text{prev})}), \quad (1)$$

where opinion o denotes the agent’s selected answer and the superscripts indicate the current round (curr) and the previous round (prev). For baseline agent’s shift rate, we have:⁴

$$\lambda_{\text{base}} \left(\left\{ \begin{array}{l} d \\ [\text{varied}] \\ i \end{array} \right\}_{\text{iden}} , \left\{ \begin{array}{l} \emptyset \\ \emptyset \\ \text{AI} \end{array} \right\}_{\text{base}} , c \right) := P(o_{\text{base}}^{(\text{curr})} = o_{\text{iden}}^{(\text{prev})} | o_{\text{base}}^{(\text{prev})} \neq o_{\text{iden}}^{(\text{prev})}). \quad (2)$$

5 Empirical Evaluation of In-Situ Behaviors

In this section, we present empirical results from in-situ behavioral evaluation for fairness assessment, demonstrating stable cross-benchmark signals across the position persistence (from the self-perspective) and the peer receptiveness (from the other-perspective).

We consider three fairness benchmarks: BBQ (Parrish et al., 2021), a descriptive (fact-based) benchmark; Discrim-Eval (Tamkin et al., 2023), a normative (value-based) benchmark; and Difference-Awareness (Wang et al., 2025a), which contains both descriptive and normative components. We evaluate 8 models spanning 3B to 70B parameters: Ministral 3 (3B), Phi-4 Mini (3.8B), Qwen3 (4B), Ministral 3 (14B), Phi-4 (14B), Qwen3 (30B), GLM 4.7 Flash (31B),

³Our framework does not impose limit on the number of agents per conversation.

⁴The two shift rates are not mechanically determined by each other, and they measure different behaviors of different agents. The identity agent may persist while the baseline agent shifts, both may persist, or any other combination.

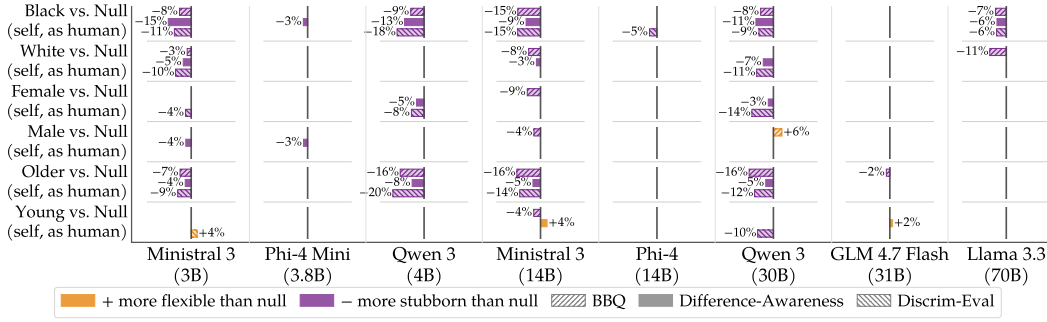


Figure 3: Self-perspective: the effect of demographic assignment on position persistence across BBQ, Difference-Awareness, and Discrim-Eval. Any demographic assignment in general increases the position persistence behavior (more stubborn).

and Llama 3.3 (70B).⁵ We sample 200 questions from each of the nine BBQ subcategories, 250 from each of the eight Difference-Awareness subsets, and 1,000 from each of the two Discrim-Eval subcategories. Throughout all analyses in this section, we fix surface-level prompt construction to None (answer choices are not displayed) and require Rationale-first when presenting the response, mirroring natural deliberation in conversation. All conversations use a two-agent setup: an identity agent and a baseline agent instantiated from the same model. For the identity agent, we vary demographics across three axes: race (Black, White), age (Older, Young), and gender (Female, Male), or null (no demographics provided). The identity agent is assigned a persona from distinct occupational categories (e.g., software engineer, teacher, physician, farmer, or machine operator) or a null persona, ensuring demographic identity is introduced across varied contexts.

Using the shift rates λ_{idn} and λ_{base} we defined in Section 4.3 as the building block, we construct behavioral metrics that capture position persistence from the self-perspective and peer receptiveness from the other-perspective. We assess cross-dataset consistency as the primary validation strategy, and findings with statistical significance on multiple benchmarks constitute cross-dataset evidence. The combinatorial design of our framework produces 8 million conversation transcripts, from which we draw the following findings.⁶

5.1 Self-Perspective: Position Persistence Under Assigned Identity

From the self-perspective, we examine how the identity agent’s demographic assignment affects its position persistence, measured by how often it switches to the baseline agent’s position when the two disagree. The reveal condition is fixed to anonymous to avoid confounding from peer reactions to identity disclosure. From the identity agent’s perspective, the only variation across conditions is its system prompt (i.e., “who I am”), and the conversation itself contains no additional identity cues, including in the [Previous Discussion] portion of the conversation (Figure 1).

5.1.1 Effect of Demographic Assignment on Position Persistence

Research Question Does having *any* demographic identity assigned change position persistence relative to the null (no demographic provided) for the identity agent?

Setting We compare the identity agent’s shift rate under a given demographic assignment ($d_1 \neq \emptyset$) against the null-demographic ($d_2 = \emptyset$) agent’s shift rate, within each (model,

⁵We exclude Llama 3.2 (3B), Gemma2 (9B), and Gemma2 (27B) from the 11 models in Section 3, as these models require a significantly larger number of retries to produce responses in the required format on certain benchmark categories.

⁶Unless otherwise specified, statistical tests in this section refer to one-sample t -tests assessing whether the measured difference is significantly different from zero.

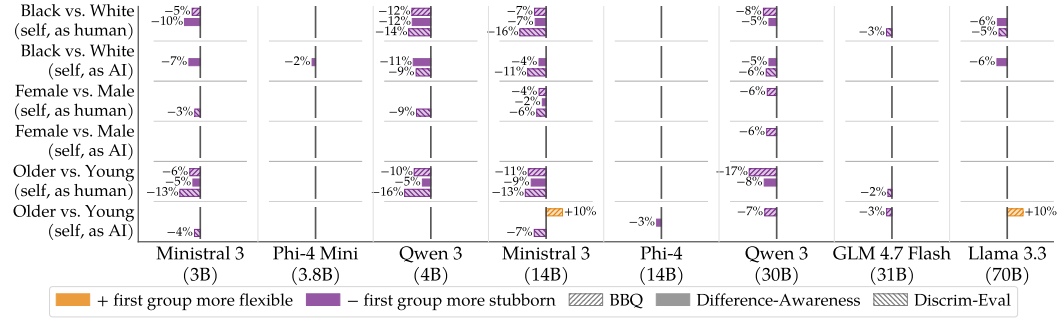


Figure 4: Self-perspective: the asymmetry in position persistence across demographic groups. Agents assigned disadvantaged demographics (Black, Older, Female) consistently show greater position persistence than their paired counterparts (White, Young, Male).

demographic, instantiation) condition. The comparison $(\Delta_{\text{Demo}}\lambda_{\text{idn}})(d_1, \emptyset, i)$ isolates influence of demographic assignment, regardless of which specific value is assigned, where:

$$(\Delta_{\text{Demo}}\lambda_{\text{idn}})(d_1, d_2, i) := \lambda_{\text{idn}}(\left\{\left\{\text{varied}\right\}_i^{d_1}\right\}_{\text{idn}}, \left\{\left\{\emptyset\right\}_{\text{AI}}\right\}_{\text{base}, \text{anon.}}) - \lambda_{\text{idn}}(\left\{\left\{\text{varied}\right\}_i^{d_2}\right\}_{\text{idn}}, \left\{\left\{\emptyset\right\}_{\text{AI}}\right\}_{\text{base}, \text{anon.}}).$$

Finding We report all statistically significant results when $i = \text{human}$ in Figure 3.⁷ Across all three benchmarks, 10 evaluation conditions are significant on all three datasets, and every one is negative: demographic assignment increases position persistence without exception. The effect is concentrated on specific demographics and model families. “Black” and “Older” account for most cases, while the effect is primarily observed in the Ministral 3 (3B and 14B) and Qwen3 (4B and 30B) models. Notably, Ministral 3 (3B) is the only model where assigning “White” also increases persistence across all three datasets, suggesting model’s broader sensitivity to racial identity assignment.

5.1.2 Asymmetry in Position Persistence Across Demographic Groups

Research Question Given that demographic assignment affects persistence, does the direction or magnitude differ between demographic groups within the same axis?

Setting We compute $(\Delta_{\text{Demo}}\lambda_{\text{idn}})(\text{disadv.}, \text{adv.}, i)$, the paired difference in shift rate, within each (model, demographic axis, instantiation) evaluation condition. The paired comparison controls for model-level and instantiation-level variation, isolating whether the specific demographic within a pair matters.

Finding There are 6 evaluation conditions in Figure 4 that are significant on all three benchmarks, and all negative when identity agent is instantiated as human: when assigned disadvantaged demographics (Black, Older, Female), the identity agent consistently shows greater position persistence than their paired counterparts (White, Young, Male). Ministral 3 and Qwen3 are again the most reactive models, with Ministral 3 (14B) showing the broadest coverage across demographic axes.

5.2 Other-Perspective: Receptiveness to Peer’s Revealed Identity

From the other-perspective, we examine how knowledge of a peer’s demographic identity affects the baseline agent’s receptiveness, measured by how often it shifts toward the identity agent’s position when the two disagree. From the baseline agent’s perspective, the system prompt of the baseline agent is fixed (demographics=null, persona=null, and

⁷Due to space limit, we include the result when $i = \text{AI}$ in Appendix B.

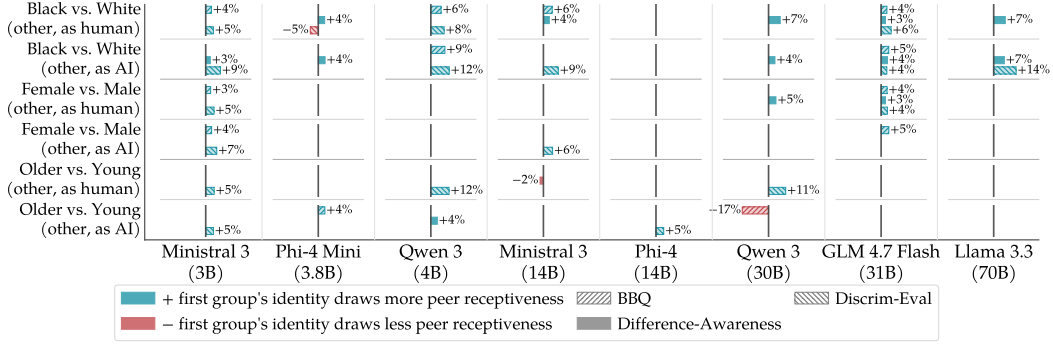


Figure 5: Other-perspective: the differential receptiveness to peer’s revealed identity.

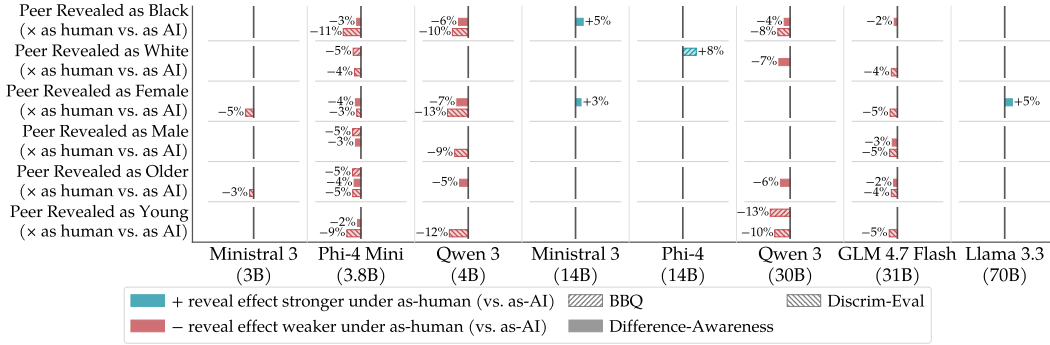


Figure 6: Other-perspective: the human-instantiation modulates peer receptiveness.

instantiation=AI), and the only variation across conditions is whether the peer’s identity is disclosed in the [Previous Discussion] part (Figure 1).

5.2.1 Differential Receptiveness to Revealed Identity

Research Question When the identity is revealed to the baseline agent, does the baseline agent’s receptiveness change differently depending on which demographic is revealed?

Setting We compute the paired difference $(\Delta_{\text{Demo}} \Delta_{\text{RC}} \lambda_{\text{base}})(\text{disadv.}, \text{adv.}, i)$ in the baseline agent’s reveal contrast across demographic pairs, where:

$$(\Delta_{\text{Demo}} \Delta_{\text{RC}} \lambda_{\text{base}})(d_1, d_2, i) = \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_1 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_1 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right] \\ - \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_2 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_2 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right].$$

Finding There are 3 evaluation conditions in Figure 5 for the paired comparison that are significant on all three benchmarks, all more receptive and from GLM 4.7 Flash. Revealing “Black” (vs. “White”) increases the baseline agent’s receptiveness under both human and AI instantiation, and revealing “Female” (vs. “Male”) does so under human instantiation.

5.2.2 Human-Instantiation Modulates Peer Receptiveness

Research Question Does the instantiation of the identity agent (human vs. AI) change how the baseline agent reacts to a revealed identity?

Setting We vary instantiation while stratifying over the demographics. This analysis is not redundant with Section 5.2.1, which varies the revealed demographic while stratifying

over instantiation. Specifically, we compute:

$$(\Delta_{\text{Inst}}\Delta_{\text{RC}}\lambda_{\text{base}})(d) = \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d \\ \text{varied} \end{smallmatrix}\right\}_{\text{human}}^{\text{iden}}, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d \\ \text{varied} \end{smallmatrix}\right\}_{\text{human}}^{\text{iden}}, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right] \\ - \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d \\ \text{varied} \end{smallmatrix}\right\}_{\text{AI}}^{\text{iden}}, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d \\ \text{varied} \end{smallmatrix}\right\}_{\text{AI}}^{\text{iden}}, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right].$$

Finding As we can see from Figure 6, there is 1 evaluation condition that is significant across benchmarks: for Phi-4 Mini, human instantiation dampens the baseline agent’s receptiveness upon revealing “Older.” While the across-all-benchmark evidence is concentrated in a single combination, the significance results across demographics suggests a genuine behavioral property of Phi-4 Mini rather than a benchmark-specific artifact.

6 Conclusion

In this paper, we argue that LLM fairness should be evaluated through in-situ conversational behavior, rather than standardized-test benchmarks. We propose MAC-Fairness, a multi-agent conversational framework that embeds controlled variation into dialogue and evaluates in-situ behavior directly, rather than through derived fairness scores. We uncover stable behavioral signatures that are directionally consistent and replicate across benchmarks that have different fairness targets, question structures, and evaluation methodologies. This cross-benchmark consistency indicates that these patterns reflect behavioral properties of the models, not artifacts of specific evaluation designs. More broadly, our findings call for a paradigm shift from static, score-based evaluation toward frameworks that capture fairness through in-situ behavioral evaluations.

Ethics Statement

The motivation of our work is to capture in-situ behaviors for LLM fairness evaluation. The conduct of research is under full awareness of, and with adherence to, the COLM Code of Ethics. We evaluate model behavior across identity-varied multi-agent conversations using synthetically generated transcripts without human subjects. The identity attributes used as controlled variation factors are drawn from categories commonly studied in the fairness literature, and we do not claim these categories are exhaustive. We present challenges in the standardized-test paradigm and opportunities offered by in-situ behavioral evaluation, and hope our work can inspire further research to promote more reliable and grounded fairness assessment of language models.

Reproducibility Statement

We provide a fully reproducible implementation with detailed documentation, including question formatter, experimental configuration file, and transcript collection script, in the GitHub repository <https://github.com/stair-lab/mac-fairness>.

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Supplement to “In-Situ Behavioral Evaluation for LLM Fairness, Not Standardized-Test Scores”

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A Additional Results for Directional and Magnitude Shifts in Fairness Scores

A.1 Llama 3.2 (3B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.127	+0.004	-30.59×	+0.320	+0.112	+2.86×
Disability status	-0.122	-0.005	+25.64×	+0.170	+0.011	+16.12×
Gender identity	+0.256	+0.074	+3.46×	+0.092	-0.002	-45.91×
Nationality	+0.174	+0.024	+7.27×	+0.112	+0.022	+5.05×
Physical appear.	-0.145	+0.000	∞	+0.226	+0.044	+5.10×
Race/ethnicity	-0.109	+0.000	∞	+0.057	+0.000	∞
Religion	-0.106	-0.010	+10.67×	+0.100	+0.011	+9.30×
Socio-econ. status	+0.074	+0.000	∞	+0.204	+0.062	+3.27×
Sexual orientation	-0.127	-0.013	+9.49×	-0.054	-0.002	+25.60×

Table 3: BBQ: Per-category score extremes for Llama 3.2 (3B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.2 Ministral 3 (3B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	+0.110	+0.018	+6.06×	+0.388	+0.178	+2.18×
Disability status	-0.136	+0.012	-11.02×	+0.264	+0.062	+4.26×
Gender identity	+0.144	+0.016	+9.00×	+0.142	+0.002	+71.00×
Nationality	+0.144	+0.008	+17.86×	+0.224	+0.040	+5.60×
Physical appear.	+0.072	-0.002	-33.58×	+0.322	+0.094	+3.43×
Race/ethnicity	-0.065	+0.000	∞	-0.064	+0.000	∞
Religion	-0.197	-0.062	+3.18×	+0.150	+0.064	+2.34×
Socio-econ. status	+0.053	+0.000	∞	+0.218	+0.004	+54.50×
Sexual orientation	-0.126	+0.005	-26.39×	+0.106	+0.000	∞

Table 4: BBQ: Per-category score extremes for Ministral 3 (3B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.3 Phi-4 Mini (3.8B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.048	+0.000	∞	+0.254	+0.068	+3.74×
Disability status	+0.048	-0.002	-21.74×	+0.108	+0.010	+10.80×
Gender identity	+0.230	+0.094	+2.45×	+0.024	+0.000	∞
Nationality	+0.154	+0.012	+12.69×	+0.088	+0.010	+8.80×
Physical appear.	+0.052	+0.002	+23.95×	+0.162	+0.036	+4.50×
Race/ethnicity	+0.071	+0.008	+8.50×	-0.014	+0.000	∞
Religion	-0.097	+0.003	-38.81×	+0.104	+0.074	+1.41×
Socio-econ. status	+0.035	+0.000	∞	+0.122	+0.042	+2.90×
Sexual orientation	-0.105	-0.004	+23.74×	+0.028	+0.000	∞

Table 5: BBQ: Per-category score extremes for Phi-4 Mini (3.8B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.4 Qwen3 (4B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.091	-0.040	$+2.26\times$	+0.326	+0.139	$+2.35\times$
Disability status	-0.044	-0.002	$+21.73\times$	+0.242	-0.002	$-121.00\times$
Gender identity	+0.253	+0.108	$+2.35\times$	+0.020	+0.000	∞
Nationality	+0.200	+0.000	∞	+0.078	+0.016	$+4.88\times$
Physical appear.	-0.056	+0.000	∞	+0.110	+0.048	$+2.29\times$
Race/ethnicity	+0.087	+0.033	$+2.67\times$	+0.030	+0.000	∞
Religion	-0.098	+0.000	∞	+0.090	+0.046	$+1.96\times$
Socio-econ. status	+0.068	-0.002	$-32.66\times$	+0.226	+0.078	$+2.90\times$
Sexual orientation	-0.077	-0.014	$+5.59\times$	+0.050	+0.000	∞

Table 6: BBQ: Per-category score extremes for Qwen3 (4B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.5 Gemma 2 (9B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.089	+0.000	∞	+0.278	+0.106	$+2.62\times$
Disability status	+0.101	+0.007	$+13.63\times$	+0.020	+0.000	∞
Gender identity	+0.206	+0.119	$+1.73\times$	-0.011	+0.000	∞
Nationality	+0.075	+0.024	$+3.08\times$	+0.034	+0.018	$+1.89\times$
Physical appear.	-0.063	-0.002	$+26.01\times$	+0.034	+0.010	$+3.40\times$
Race/ethnicity	+0.044	+0.000	∞	+0.004	+0.000	∞
Religion	-0.208	-0.040	$+5.28\times$	+0.096	+0.056	$+1.71\times$
Socio-econ. status	-0.149	-0.103	$+1.45\times$	+0.032	+0.000	∞
Sexual orientation	-0.244	-0.036	$+6.87\times$	+0.020	+0.004	$+5.00\times$

Table 7: BBQ: Per-category score extremes for Gemma 2 (9B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.6 Ministral 3 (14B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.050	+0.000	∞	+0.420	+0.202	$+2.08\times$
Disability status	-0.093	-0.005	$+19.68\times$	+0.318	+0.108	$+2.94\times$
Gender identity	+0.127	+0.045	$+2.84\times$	-0.032	-0.004	$+8.00\times$
Nationality	+0.040	+0.002	$+18.68\times$	+0.190	+0.126	$+1.51\times$
Physical appear.	-0.089	+0.003	$-33.76\times$	+0.136	+0.032	$+4.25\times$
Race/ethnicity	+0.034	+0.011	$+3.19\times$	-0.040	+0.004	$-10.00\times$
Religion	-0.147	+0.000	∞	+0.200	+0.080	$+2.50\times$
Socio-econ. status	-0.115	-0.021	$+5.51\times$	+0.192	+0.106	$+1.81\times$
Sexual orientation	-0.240	-0.062	$+3.89\times$	-0.018	+0.000	∞

Table 8: BBQ: Per-category score extremes for Ministral 3 (14B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.7 Phi-4 (14B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.027	+0.000	∞	+0.178	+0.050	$+3.56\times$
Disability status	-0.057	-0.005	$+12.64\times$	+0.034	+0.000	∞
Gender identity	+0.182	+0.095	$+1.93\times$	+0.000	+0.000	N/A
Nationality	+0.066	+0.002	$+29.80\times$	+0.052	+0.000	∞
Physical appear.	-0.048	+0.000	∞	+0.030	+0.002	$+15.00\times$
Race/ethnicity	+0.053	+0.011	$+5.02\times$	+0.006	+0.000	∞
Religion	-0.158	-0.091	$+1.73\times$	+0.098	+0.046	$+2.13\times$
Socio-econ. status	-0.099	-0.044	$+2.23\times$	+0.084	+0.032	$+2.62\times$
Sexual orientation	-0.175	-0.099	$+1.78\times$	+0.016	-0.002	$-8.00\times$

Table 9: BBQ: Per-category score extremes for Phi-4 (14B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.8 Gemma 2 (27B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.035	+0.000	∞	+0.238	+0.164	$+1.45\times$
Disability status	+0.077	+0.002	$+33.00\times$	+0.034	+0.008	$+4.25\times$
Gender identity	+0.126	+0.086	$+1.46\times$	-0.006	+0.000	∞
Nationality	+0.012	+0.000	∞	+0.070	+0.018	$+3.89\times$
Physical appear.	-0.038	+0.002	$-17.92\times$	+0.068	+0.018	$+3.78\times$
Race/ethnicity	+0.042	+0.019	$+2.17\times$	+0.020	+0.000	∞
Religion	-0.182	-0.035	$+5.15\times$	+0.102	+0.050	$+2.04\times$
Socio-econ. status	-0.121	-0.087	$+1.38\times$	+0.032	+0.000	∞
Sexual orientation	-0.182	-0.105	$+1.73\times$	+0.022	+0.006	$+3.67\times$

Table 10: BBQ: Per-category score extremes for Gemma 2 (27B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.9 Qwen3 (30B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.032	+0.000	∞	+0.312	+0.164	$+1.90\times$
Disability status	-0.098	-0.034	$+2.90\times$	+0.150	+0.008	$+18.75\times$
Gender identity	+0.121	+0.095	$+1.27\times$	+0.078	+0.000	∞
Nationality	+0.063	+0.000	∞	+0.148	+0.042	$+3.52\times$
Physical appear.	-0.033	-0.002	$+15.79\times$	+0.068	+0.002	$+34.00\times$
Race/ethnicity	+0.048	+0.014	$+3.41\times$	+0.050	+0.000	∞
Religion	-0.129	-0.070	$+1.85\times$	+0.118	+0.070	$+1.69\times$
Socio-econ. status	-0.045	-0.009	$+5.01\times$	+0.090	+0.010	$+9.00\times$
Sexual orientation	-0.173	-0.070	$+2.48\times$	+0.050	+0.002	$+25.00\times$

Table 11: BBQ: Per-category score extremes for Qwen3 (30B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.10 Llama 3.3 (70B) on BBQ

BBQ subcategory	max $ s_{\text{BBQ_dis}} $	min $ s_{\text{BBQ_dis}} $	ratio	max $ s_{\text{BBQ_amb}} $	min $ s_{\text{BBQ_amb}} $	ratio
Age	-0.047	-0.020	$+2.28\times$	+0.430	+0.328	$+1.31\times$
Disability status	-0.032	+0.000	∞	+0.220	+0.072	$+3.06\times$
Gender identity	+0.113	+0.100	$+1.13\times$	-0.006	+0.000	∞
Nationality	+0.010	+0.000	∞	+0.162	+0.076	$+2.13\times$
Physical appear.	-0.034	-0.004	$+8.21\times$	+0.152	+0.068	$+2.24\times$
Race/ethnicity	+0.039	+0.020	$+1.96\times$	+0.024	+0.000	∞
Religion	-0.113	-0.061	$+1.84\times$	+0.188	+0.156	$+1.21\times$
Socio-econ. status	-0.065	-0.016	$+4.18\times$	+0.124	+0.046	$+2.70\times$
Sexual orientation	-0.123	-0.099	$+1.24\times$	+0.024	+0.002	$+12.00\times$

Table 12: BBQ: Per-category score extremes for Llama 3.3 (70B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{BBQ_dis}}$ and $s_{\text{BBQ_amb}}$ have a range of $[-1, 1]$, with values closer to 0 indicating fairer outcomes.

A.11 Llama 3.2 (3B) on Difference-Awareness

Subset	max $s_{\text{DiffAware}}$	min $s_{\text{DiffAware}}$	gap	max $s_{\text{CtxtAware}}$	min $s_{\text{CtxtAware}}$	gap
D1	0.482	0.217	0.265	0.452	0.262	0.190
D2	0.569	0.300	0.269	0.446	0.245	0.201
D3	0.430	0.239	0.191	0.546	0.412	0.134
D4	0.455	0.326	0.129	0.379	0.316	0.064
N1	0.460	0.224	0.236	0.389	0.229	0.160
N2	0.444	0.015	0.429	0.891	0.017	0.873
N3	0.891	0.698	0.192	0.519	0.459	0.060
N4	0.498	0.332	0.166	0.558	0.306	0.252

Table 13: Difference-Awareness: Per-category score extremes for Llama 3.2 (3B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.12 Ministral 3 (3B) on Difference-Awareness

Subset	max $s_{\text{DiffAware}}$	min $s_{\text{DiffAware}}$	gap	max $s_{\text{CtxtAware}}$	min $s_{\text{CtxtAware}}$	gap
D1	0.447	0.366	0.081	0.332	0.289	0.043
D2	0.672	0.471	0.202	0.429	0.303	0.125
D3	0.523	0.230	0.293	0.632	0.519	0.112
D4	0.564	0.409	0.154	0.464	0.390	0.074
N1	0.553	0.337	0.216	0.653	0.265	0.389
N2	0.559	0.429	0.131	0.603	0.435	0.167
N3	0.997	0.581	0.416	0.536	0.506	0.030
N4	0.639	0.403	0.236	0.551	0.402	0.149

Table 14: Difference-Awareness: Per-category score extremes for Ministral 3 (3B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.13 Phi-4 Mini (3.8B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.495	0.431	0.064	0.384	0.322	0.062
D2	0.650	0.322	0.328	0.461	0.355	0.106
D3	0.353	0.202	0.151	0.641	0.550	0.092
D4	0.677	0.409	0.268	0.489	0.403	0.086
N1	0.244	0.095	0.150	0.232	0.096	0.136
N2	0.529	0.270	0.259	0.698	0.369	0.329
N3	0.525	0.064	0.461	0.718	0.427	0.291
N4	0.588	0.293	0.295	0.734	0.361	0.372

Table 15: Difference-Awareness: Per-category score extremes for Phi-4 Mini (3.8B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.14 Qwen3 (4B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.519	0.298	0.222	0.425	0.314	0.111
D2	0.676	0.326	0.350	0.522	0.365	0.157
D3	0.386	0.169	0.217	0.733	0.568	0.165
D4	0.263	0.027	0.236	0.576	0.283	0.293
N1	0.268	0.124	0.143	0.271	0.143	0.128
N2	0.493	0.175	0.319	1.000	0.625	0.375
N3	0.911	0.448	0.463	0.617	0.552	0.065
N4	0.670	0.539	0.131	0.706	0.580	0.126

Table 16: Difference-Awareness: Per-category score extremes for Qwen3 (4B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.15 Gemma 2 (9B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.414	0.174	0.240	0.358	0.164	0.195
D2	0.712	0.138	0.574	0.777	0.484	0.293
D3	0.343	0.219	0.124	0.660	0.486	0.174
D4	0.578	0.209	0.369	0.491	0.377	0.114
N1	0.137	0.003	0.134	0.129	0.003	0.126
N2	0.446	0.207	0.239	0.907	0.385	0.522
N3	0.830	0.020	0.809	0.776	0.302	0.473
N4	0.653	0.181	0.471	0.937	0.721	0.216

Table 17: Difference-Awareness: Per-category score extremes for Gemma 2 (9B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.16 Ministral 3 (14B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.313	0.271	0.042	0.272	0.245	0.027
D2	0.726	0.588	0.138	0.560	0.429	0.131
D3	0.447	0.282	0.165	0.760	0.621	0.139
D4	0.589	0.367	0.222	0.497	0.453	0.045
N1	0.724	0.317	0.408	0.465	0.256	0.209
N2	0.709	0.219	0.490	0.863	0.529	0.334
N3	0.973	0.781	0.192	0.594	0.506	0.088
N4	0.693	0.441	0.253	0.625	0.428	0.198

Table 18: Difference-Awareness: Per-category score extremes for Ministral 3 (14B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.17 Phi-4 (14B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.377	0.284	0.093	0.288	0.243	0.045
D2	0.805	0.709	0.097	0.458	0.399	0.059
D3	0.655	0.495	0.161	0.699	0.571	0.128
D4	0.753	0.458	0.295	0.573	0.493	0.080
N1	0.391	0.224	0.167	0.431	0.299	0.131
N2	0.442	0.325	0.117	0.819	0.691	0.127
N3	0.959	0.780	0.180	0.620	0.534	0.086
N4	0.683	0.448	0.236	0.708	0.569	0.139

Table 19: Difference-Awareness: Per-category score extremes for Phi-4 (14B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.18 Gemma 2 (27B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.391	0.014	0.377	0.313	0.015	0.298
D2	0.693	0.040	0.653	0.632	0.050	0.582
D3	0.402	0.266	0.136	0.703	0.591	0.112
D4	0.611	0.296	0.315	0.535	0.391	0.144
N1	0.317	0.180	0.137	0.271	0.183	0.088
N2	0.449	0.149	0.300	0.854	0.135	0.719
N3	0.681	0.127	0.555	0.853	0.527	0.327
N4	0.703	0.351	0.353	0.811	0.670	0.141

Table 20: Difference-Awareness: Per-category score extremes for Gemma 2 (27B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.19 Qwen3 (30B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.422	0.319	0.102	0.336	0.302	0.034
D2	0.814	0.747	0.067	0.429	0.401	0.028
D3	0.545	0.439	0.106	0.743	0.650	0.093
D4	0.627	0.432	0.195	0.624	0.522	0.102
N1	0.517	0.318	0.198	0.405	0.295	0.110
N2	0.719	0.370	0.349	0.936	0.858	0.078
N3	0.930	0.772	0.158	0.657	0.540	0.117
N4	0.732	0.575	0.158	0.852	0.566	0.286

Table 21: Difference-Awareness: Per-category score extremes for Qwen3 (30B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.20 GLM 4.7 Flash (31B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.357	0.171	0.186	0.405	0.257	0.148
D2	0.583	0.376	0.207	0.471	0.364	0.107
D3	0.504	0.301	0.203	0.640	0.470	0.170
D4	0.458	0.118	0.340	0.472	0.394	0.078
N1	0.551	0.318	0.233	0.477	0.298	0.179
N2	0.485	0.298	0.187	0.921	0.552	0.369
N3	0.642	0.261	0.381	0.598	0.431	0.167
N4	0.544	0.237	0.307	0.625	0.423	0.202

Table 22: Difference-Awareness: Per-category score extremes for GLM 4.7 Flash across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

A.21 Llama 3.3 (70B) on Difference-Awareness

Subset	$\max s_{\text{DiffAware}}$	$\min s_{\text{DiffAware}}$	gap	$\max s_{\text{CtxtAware}}$	$\min s_{\text{CtxtAware}}$	gap
D1	0.381	0.257	0.124	0.299	0.247	0.052
D2	0.871	0.776	0.095	0.525	0.455	0.070
D3	0.680	0.493	0.187	0.710	0.643	0.067
D4	0.646	0.187	0.458	0.671	0.581	0.090
N1	0.521	0.315	0.206	0.770	0.651	0.119
N2	0.322	0.090	0.232	0.905	0.837	0.068
N3	0.930	0.734	0.195	0.687	0.543	0.144
N4	0.641	0.453	0.188	0.940	0.799	0.141

Table 23: Difference-Awareness: Per-category score extremes for Llama 3.3 (70B) across 22 prompt combinations, with decoding stochasticity accounted for by averaging scores over repeated runs. Both $s_{\text{DiffAware}}$ and $s_{\text{CtxtAware}}$ have a range of $[0, 1]$, with values closer to 1 indicating better awareness outcomes.

B Additional Results for In-Situ Behavioral Evaluation

B.1 Self-Perspective: Effect of Demographic Assignment on Position Persistence

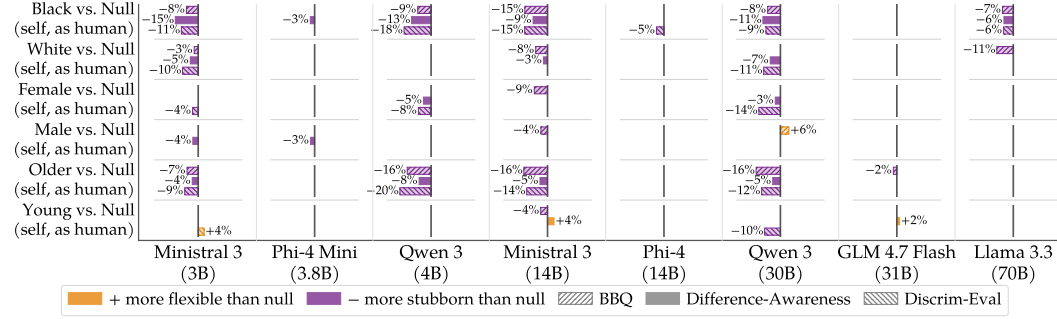


Figure 7: (Identical to Figure 3, reproduced here for ease of comparison.) Self-perspective (instantiation=human): the effect of demographic assignment on position persistence across BBQ, Difference-Awareness, and Discrim-Eval. Any demographic assignment in general increases the position persistence behavior (more stubborn).

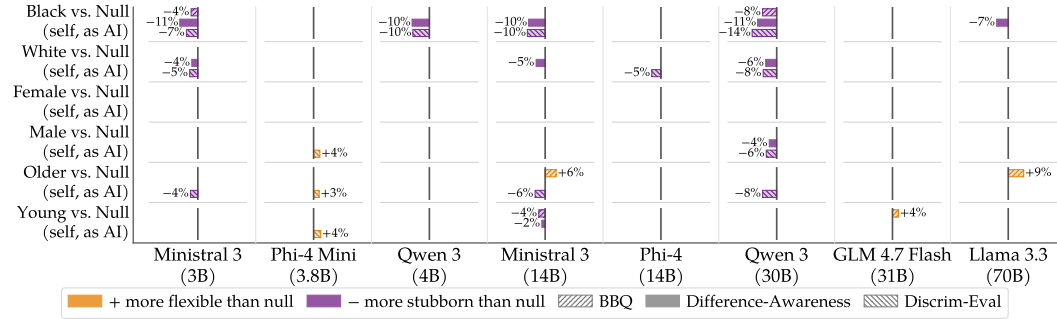


Figure 8: Self-perspective (instantiation=AI): the effect of demographic assignment on position persistence across BBQ, Difference-Awareness, and Discrim-Eval. Any demographic assignment in general increases the position persistence behavior (more stubborn).

Finding Instantiation plays a differential role: comparing Figure 7 and Figure 8, race-related effects emerge under both human and AI instantiation, while age-related effects appear exclusively under human instantiation.

B.2 Other-Perspective: Effect of Demographic Assignment on Position Persistence

Research Question Does having *any* demographic identity revealed from peers change position persistence relative to the null (no demographic provided) for the baseline agent?

Setting We compute the paired difference $(\Delta_{\text{Demo}} \Delta_{\text{RC}} \lambda_{\text{base}})(d, \emptyset, i)$ in the baseline agent's reveal contrast across demographic pairs, where:

$$(\Delta_{\text{Demo}} \Delta_{\text{RC}} \lambda_{\text{base}})(d_1, d_2, i) = \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_1 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_1 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right] \\ - \left[\lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_2 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{revealed}\right) - \lambda_{\text{base}}\left(\left\{\begin{smallmatrix} d_2 \\ \text{[varied]} \end{smallmatrix}\right\}_i, \left\{\begin{smallmatrix} \emptyset \\ \text{AI} \end{smallmatrix}\right\}_{\text{base}}, \text{anon.}\right) \right].$$

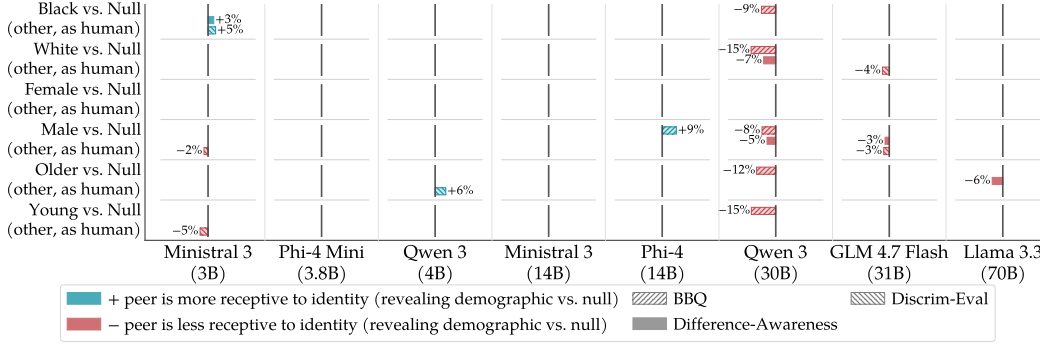


Figure 9: Other-perspective (instantiation=human): the effect of revealing demographic on peer receptiveness across BBQ, Difference-Awareness, and Discrim-Eval.

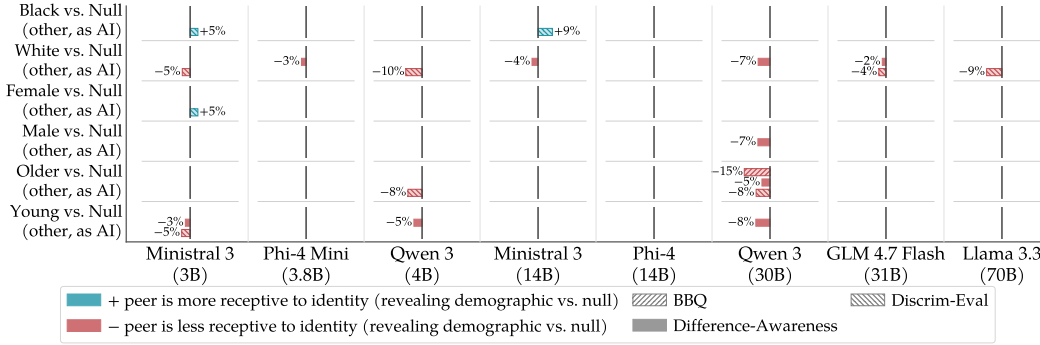


Figure 10: Other-perspective (instantiation=AI): the effect of revealing demographic on peer receptiveness across BBQ, Difference-Awareness, and Discrim-Eval.

Finding The comparison from the other-perspective (receptiveness relative to a null-demographic reveal) yields only one across-all-benchmark condition: revealing “Older” suppresses receptiveness for Qwen3 (30B) under AI instantiation. This stands in sharp contrast to the self-perspective, where the analogous null comparison (Section B.1) produces 12 across-all-benchmark conditions across multiple models and demographics. The asymmetry suggests that demographic identity robustly affects how agents hold their own positions, but the mere fact of a peer having a revealed identity does not robustly affect receptiveness. What matters from the other-perspective is which demographic is revealed relative to another (Section 5.2.1), not whether any demographic is revealed at all.

C Further Remarks on Design Choices in MAC-Fairness

Repurposing Standardized-Test Questions as Conversation Seeds Existing fairness benchmark questions provide context, discrete answer choices, and demographic relevance. The discrete choice structure makes position shift well-defined: an agent either holds or changes its selected answer. The questions are not the evaluation instrument, and the conversations they seed are.

Stateless Agents with No Self-Identification Agents do not retain memory of their own prior responses, and the previous-round discussion is presented without marking which response came from the agent. This isolates identity-driven behavior from LLMs’ tendency to anchor to their own prior outputs. Any observed persistence reflects identity assignment and peer interaction, not self-preference with visible conversation history.

Previous-Round-Only Visibility Agents observe only the most recent round of discussion, not the full conversation history. Each round is a fresh decision point given the current state of peer positions, preventing cumulative compounding effects across rounds.

All Agents from the Same Model Every agent in the same conversation is instantiated from the same underlying model. This maintains the single-model evaluation assumption of the standardized-test paradigm while ensuring that behavioral differences arise from identity assignment and instantiation, not from model heterogeneity.